Can one local reading pass Cassini and keep a wide-binary signal? The QUMOND chain of MM 2.56 at 17σ, and a reading registered for Gaia DR4

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Abstract

Cassini ranging data bound the external-field quadrupole of the Solar System at $Q_2 = (1.6 \pm 1.8)\cdot 10^{-27}$ s^{-2} (Park et al. 2026). The wide-binary chain registered in MM 2.56 has these ingredients: Taken down to the Sun with the same field equation, it predicts $Q_2 = 3.22\cdot 10^{-26}$ s^{-2}, 17σ above the measurement (17.0–19.4σ over eight input choices; the MM 2.56 solver gives the same number within $3\cdot 10^{-5}$). A local reading of the external field. Here the anomaly lives in the relation between two bodies. What it gives. Registered for Gaia DR4. The reading has its own orbit library, the unchanged forward model of MM 2.56, and a decision rule against Newton (threshold ±5 in log-likelihood, subsets ±2). The copied evaluation reproduces the MM 2.56 dry run exactly. A draft protocol with three addenda registers the anisotropy test:

Can one local reading pass Cassini and keep a wide-binary signal?

Student version of MM record 2.60 (DOI 10.5281/zenodo.23533882), written with an AI assistant from the archived text; the archived text remains on Zenodo.

The archived text is in English.

Version 2.60 of the Mathematical Materiality (MM) series, October 2026.

Mathematical Materiality, MM for short, is the author's framework of ideas. Each work in the series has a number.

- This is a *result record*. A record is a work archived online, in a public archive. This record reports results.
- It also contains a *reading* of the *external field*, registered for future data (section 5). We will see shortly what the two words in italics mean.
- *Registered* means written and fixed with its date, before seeing the data. So it cannot be changed afterwards.
- Gaia is a satellite. It measures the position and the motion of a great many stars. Its data come out in batches, called data releases.
- The record continues MM 2.56 (DOI 10.5281/zenodo.23506401). A DOI is a code that points to a document online for ever.
- MM 2.56 registered a test on wide binaries. MM 2.56 and its *protocol* are unchanged. A protocol is the list of the steps of a test, written before doing it.
- The reference texts of the work are in Italian. The archived text, in English, summarises them.

How this text is built. First comes a summary of the whole record. Then sections 1 to 7 go into the details. At the end you find how the work was done. Then there is a glossary, that is, a list of the difficult words with their meaning.

In short

This section summarises the whole record. There are many new words. We say what they mean one by one, the first time they appear. You also find them again in the glossary at the end.

The Cassini measurement

The Solar System lies inside a galaxy, the Milky Way. The gravity of the Galaxy pulls the Sun and the planets. This gravity, which comes from outside, is called the *external field*. A *field* says how much and in which direction gravity pulls, at every point of space.

There is an idea in physics called **MOND**. We introduce it shortly. In MOND the external field changes the Sun's gravity a little. It stretches it along the direction of the Galactic centre. And it squeezes it at the sides. A deformation of this kind is called a **quadrupole**.

An everyday example: the tides. The Moon stretches the Earth's oceans along the line that goes from the Earth to the Moon. So the water rises on two opposite sides of the Earth. This deformation is a quadrupole too.

Cassini was a space probe. For many years it went round Saturn. From Earth it was followed with radio signals. This is called *radio tracking*. In this way Saturn's orbit is known very well. A quadrupole around the Sun would change that orbit a little.

The Cassini measurement puts a limit on the quadrupole. Its value is called Q_2 (Q-two):

$Q_2 = (1.6 \pm 1.8)\cdot 10^{-27}\ s^{-2}$

- The sign ± (plus or minus) gives the error of the measurement. The true value probably lies between 1.6 minus 1.8 and 1.6 plus 1.8. Zero is inside this band.
- 10^{-27} is a tiny number. It means one divided by a one followed by twenty-seven zeros.
- s^{-2} means divided by a second squared.
- The measurement is by Park and colleagues (Park et al. 2026, in the journal PRD; arXiv:2602.17884). *Et al.* is Latin and means *and others*. arXiv is the site where physicists put their papers online.

The QUMOND chain taken to the Sun

First a little history. In galaxies the stars far from the centre go round faster than expected. Visible matter alone, with Newton's law, is not enough to keep them in orbit so fast. There are two kinds of answer:

- add an invisible matter, **dark matter**;
- change the law of gravity when the acceleration is very small.

Acceleration says how fast the speed changes. **MOND** is the second answer. It is an idea of the physicist Milgrom. In MOND there is a special, tiny acceleration: **a_0** (a-zero). When Newton's acceleration is much larger than a_0, Newton holds. When it is smaller, gravity is stronger than Newton's.

QUMOND is a version of MOND written with a *field equation*. A field equation computes gravity at every point of space, even in empty space.

MM 2.56 had built a *chain*. It is a row of four choices, linked like the rings of a chain.

1. **$a_0 = cH_0/6$.** Here c is the speed of light. H_0 (H-zero) is the *Hubble constant*: it says how fast the universe is expanding today. In words: a_0 is the speed of light times the Hubble constant, divided by six.
2. **The law ν(y) = 1/(1 − e^(−√y)).** The letter ν (nu) gives the *boost*. The boost is the number by which Newton's gravity is multiplied. The letter y is Newton's acceleration divided by a_0. The letter e is Euler's number, a constant of mathematics. In words: when y is large, ν is almost 1, and Newton holds. When y is small, ν grows.
3. **The Milky-Way field, V^2/R.** V is the speed at which the Sun goes round the centre of the Galaxy. R is the Sun's distance from the centre. In words: the speed squared divided by the distance. It is the acceleration that keeps the Sun on its orbit around the Galaxy.
4. **QUMOND**, with its field equation.

Taken down to the Sun, the chain predicts:

$Q_2 = 3.22\cdot 10^{-26}\ s^{-2}$

- This value lies 17σ above the measurement. The letter σ (sigma) stands for the typical error of the measurement. 17σ means a distance of seventeen typical errors. It is a very large distance.
- The starting choices can be made in eight different ways. In all of them the distance goes from 17.0σ to 19.4σ.

An example with numbers chosen by us. Someone predicts that an empty scale will show 17 grams. You look at it: it shows zero. The scale is usually off by 1 gram, up or down. The prediction then lies 17 typical errors away from the measurement: 17σ. That prediction does not hold.

The outcome. Cassini excludes the *global-field reading* down to the Sun. It is the reading of the chain: the field of the Galaxy enters all of space through the field equation, down to the Sun. This reading falls. The outcome hits this reading, not the MM framework.

The local reading (section 4)

The record proposes another reading: the **local reading**. First we need three words.

- **Read, reading.** In MM a body is said to *read* a field when its motion takes it into account. A *reading* is a rule of this kind. It is an interpretation, written so that it can be put to the test.
- **Anomaly.** It is the part of gravity in excess of Newton.
- **MCD.** It is the *greatest common divisor*. At school it is used with numbers. An example: the MCD of 6 and 9 is 3. In the MM framework the MCD is the common unit that a relation counts. In this record the local MCD of a mass is a precise number: **R_s/r**.
- R_s is the *Schwarzschild radius* of the mass. It is the radius that a black hole with that mass would have.
- r is the distance from the mass.
- In words: the local MCD is large near a large mass. And it decreases as one moves away.

In the local reading the anomaly does not live in empty space. It lives in the *relation* between two bodies. A relation is the link between two bodies that attract each other, like the Sun and a planet. The reading has six rules:

1. **Who reads the external field.** The body whose local MCD *prevails*, that is, is larger, reads the external field for its system.
2. **The dominant compact body.** A body is *dominant* when it prevails. It is *compact* when it is small compared with the distance of the relation. A dominant compact body reads the relation itself.
3. **Without a dominant body.** If no body prevails, the relation is read along its orbit.
4. **Interior and exterior.** The interior field and the exterior field add as magnitudes. That is, their values are added, without looking at the directions.
5. **The direction.** It enters through a *tensorial alignment*: what counts is the angle between the pair of stars and the centre of the Galaxy. How much it counts is said by the data: it is a measured amount.
6. **Light.** Light reads the *curvature* and not the external field.

We look at the rules one by one in section 4.

What the local reading gives

With these rules the local reading gives these results.

- **Saturn:** $Q_2 \approx 10^{-30}\ s^{-2}$. The sign ≈ means about equal. 10^{-30} is much smaller than 10^{-27}, the order of the Cassini measurement.
- **Planets and comets, with any orbit:** they are *Newtonian*. It means that they follow Newton's law, with no boost.
- **The star S2.** It goes round Sgr A*, the black hole at the centre of the Milky Way. For S2 there is no anomalous boost. *General relativity* remains, Einstein's theory of gravity.
- **Wide binaries.** They are pairs of stars that go round each other, very far apart. MM 2.56 had chosen 919 of them at low acceleration. For them the local reading gives a boost $\langle \eta \rangle = 1.27$. The letter η (eta) stands for the boost; the brackets $\langle\ \rangle$ stand for the mean. The QUMOND chain gives 1.32. Newton gives 1, that is, no boost.
- **Disc galaxies.** They are flat galaxies that rotate, like the Milky Way. For them the *radial acceleration relation* stays unchanged. It is an observed law. It links the measured acceleration of stars and gas to the acceleration that visible matter alone would give. *Radial* means along the radius, towards the centre.
- **The environment field.** Each galaxy also feels the gravity of nearby galaxies. It is its *environment field*. In 78 galaxies of the SPARC catalogue this field enters in full, as in the binaries. SPARC is a catalogue of galaxies whose rotation is well known.
- **Weak lensing.** Gravity bends light, like a lens. So a nearby galaxy slightly deforms the images of the distant galaxies behind it. When the deformation is small one speaks of *weak lensing*. KiDS-1000 is a large survey of the sky that measures it. Here χ^2 is 219.9 on 60 points, against 1,687.8 for Newton. χ^2 (chi-squared) measures how far a model is from the data: the smaller it is, the better.
- **Galaxies in denser environments.** They are galaxies with many other galaxies nearby. Around them light reads two to three times more at large radii, that is, far from the centre. The comparison is between all the galaxies that

act as lenses and the isolated ones.
- **Red and blue.** A *central* galaxy does not go round another, larger one. Among isolated central galaxies, red and blue ones differ by a nearly constant ratio. So the excess is compatible with the *satellites* of the full sample, rather than with a reading of light that grows with density. Satellites are small galaxies that go round a larger galaxy.

Systems without rotation (section 4.9)

In some systems the stars do not go round in order, as in a disc. They move in all directions, like the bees of a swarm. The record looks at two kinds of such systems:

- *dwarf spheroidal galaxies*: small ball-shaped galaxies;
- *globular clusters*: balls of old stars, much smaller than a galaxy.

There are three results.

- The *classical* dwarf spheroidals, that is, the best known, agree within ±0.15 dex. *Agree* means that prediction and measurement match.
- The *ultra-faint* dwarfs are very small and dim. They move 2.5 to 20 times faster than the reading predicts.
- Distant globular clusters are Newtonian.

The *dex* measures ratios in powers of ten. 1 dex means ten times. 0 dex means equal. An example with numbers chosen by us: 0.30 dex means about double.

At the same gravitational state. Here the *gravitational state* is the number x_h + e_N. x_h is the interior gravity of the system, and e_N is the external field. Both are divided by a_0. We look at them more closely in section 4.9.

- The systems with x_h + e_N between 0.019 and 0.12 are taken. They are 19 dwarfs and 11 clusters.
- The galaxies carry the boost: they sit +1.0 dex above Newton.
- The clusters do not: they sit at +0.02.

The mirror. The text speaks of the *dark-matter mirror*: in the MM framework it is the name of what is usually attributed to dark matter. A system *carries a mirror* when the mass obtained from the motion of its stars is much larger than the visible mass (section 4.9).

- The systems whose stars belong to different *generations*, that is, were born at different times, carry a mirror. This is seen from the *metallicity spread*.
- *Metallicity* is the amount of elements heavier than helium in a star. *Spread* means that the value changes from star to star.
- Single-generation clusters do not carry a mirror.

How much mirror. The amount follows a common profile, set by the radius: M_dyn ∝ r_½^1.54±0.11.

- M_dyn is the *dynamical mass*: the mass obtained from the motion of the stars.
- r_½ is the radius that contains half of the light of the system.
- The sign ∝ means *proportional to*. In words: the dynamical mass grows as the radius raised to the power 1.54, with an error of 0.11.
- The profile holds for satellites and for isolated dwarfs alike.
- Around the profile there is an *intrinsic scatter* of 0.48 dex. Intrinsic means that it does not come from the measurement errors.

Fixed before the new data (section 5)

For the next Gaia data release two checks are registered.

- **The boost.** It has its own *orbit library*, the *forward model* of MM 2.56 and a *decision rule* against Newton.
- The library is a large set of orbits computed on a computer.
- The forward model starts from the law of gravity. Then it computes what should be seen in the data.
- The decision rule says in advance when a hypothesis wins.
- **The anisotropy**, in a draft protocol with an addendum. Anisotropy means that an effect changes with direction.

The tests on today's data

On DR3. DR3 is the Gaia data release already published: these are *data already seen*. In the registered simulation series, the local reading has the highest *likelihood* of the seven hypotheses compared. The likelihood says how probable a hypothesis makes the observed data. Its logarithm is used, the *log-likelihood*. The differences are:

- +16.1 over Newton;
- +6.1 over MOND simple, one of the forms of MOND;
- +8.5 over the QUMOND chain.

The simulation noise, measured (section 5.3). Computer simulations use random numbers. The *seed* is the starting number of these random numbers. When the seed changes, the result changes a little. This change is called *noise*.

An everyday example. You shuffle a deck of cards always in the same way, always starting from the same order. You get the same sequence every time. The seed does the same for the computer's random numbers.

The noise was measured on four series of simulations, with new seeds and new libraries.

- The local reading over Newton gives +11.0 on average, from +8.2 to +16.1. It is above the threshold +5 in every series.
- The QUMOND chain over Newton ranges from −0.9 to +12.1.
- The local reading over the chain ranges from −2.0 to +9.0.

For the next release a rule over several series is registered.

1. What was already known and what is new

This section separates the work done by others from the new work of this record.

What was already known.

- **The external-field quadrupole in MOND.** It was studied by Milgrom (2009) and Blanchet and Novak (2011), in *modified-Poisson* MOND. The Poisson equation is the field equation of Newton's gravity. In this form of MOND the equation is changed.
- **The Cassini constraints.** A *constraint* is a limit set by the measurements. They were written about by Hees et al. (2014 and 2016) and Park et al. (2026).
- **Cassini and wide binaries.** AQUAL is another version of MOND with a field equation. Within QUMOND and AQUAL, Cassini implies a Newtonian wide-binary test. In words: in these theories, if Cassini does not see the quadrupole, wide binaries must turn out Newtonian too. This was shown by Desmond, Hees and Famaey (2024), Desmond (2025) and Park et al. (2026).
- **Comets.** *Long-period* comets take a very long time to go once round the Sun. These comets disfavour AQUAL: Vokrouhlický, Nesvorný and Tremaine (2024). *Disfavour* means that the data go against it, without excluding it completely.
- **Orbit-dependent readings.** They are also called *modified-inertia* readings. *Inertia* is a body's resistance to changing its speed. These readings are Milgrom's.
- **The external field in galaxies.** The external-field effect in SPARC rotation curves: Chae et al. (2020, 2021). A *rotation curve* says how fast stars and gas go round at each distance from the centre of the galaxy.
- **Weak lensing.** The radial acceleration relation measured with weak lensing: Brouwer et al. (2021).

What is new here.

- **The chain's number.** It is the value of Q_2 predicted by the chain. It is obtained with no input taken from the binaries. The *solver* of MM 2.56, that is, the program that solves the equations, reproduces it within $3 \cdot 10^{-5}$.
- **The pairs, one by one.** The 919 pairs are located one by one in the region where the quadrupole is generated.
- **One local reading** holds together four things:
- Saturn, comets and S2;
- wide binaries;
- disc galaxies in their environment;
- weak lensing.
- **A check of dwarf galaxies and distant globular clusters**, with the same reading.
- The two groups split: the galaxies are boosted, the clusters are Newtonian.
- Who carries the mirror and how much are measured. The systems with stars of different generations carry it. The

amount follows a common profile set by the radius.
- Its profile for DR4 is registered.
- **A dipole statistic for wide binaries.** The systematic effects of the sample do not reach it. A *systematic effect* is an error that always goes the same way, like a badly calibrated scale. We look at the dipole in section 4.6.

2. The data used

Here are the measured data used in the record, one by one, with their source.

The quadrupole Q_2.

- Its effect on gravity is written $\delta\Phi = -(Q_2/2)[(\hat{e}\cdot x)^2 - r^2/3]$.
- $\delta\Phi$ (delta phi) is the small change of the *gravitational potential*. The potential says how much gravitational energy each kilogram of matter has at a point.
- $\hat{e}$ is the direction towards the Galactic centre. x is the position and r the distance from the Sun.
- In words: the effect depends on how much the position is aligned with the direction of the centre.
- The value is $(1.6 \pm 1.8)\cdot 10^{-27}$ s^{-2}. Source: Park et al. 2026, equation 6, table 1.

The Solar-System acceleration.

- It is $(2.32 \pm 0.16)\cdot 10^{-10}$ m s^{-2}. It is the acceleration with which the whole Solar System falls towards the Galactic centre.
- m s^{-2} means metres per second squared. It is the unit of accelerations.
- Source: Gaia EDR3, Klioner et al. 2021. EDR3 is a first part of Gaia's third data release.

The Milky-Way field used by the chain.

- $V^2/R = 2.1456\cdot 10^{-10}$ m s^{-2}, with V = 233 km/s and R = 8.2 kpc.
- The *parsec* (pc) is a unit of distance used for stars. A kiloparsec (kpc) is a thousand parsecs.
- Source: MM 2.56.

The scale $a_0 = cH_0/6$.

- With $H_0 = 67.4$ it is $1.0914\cdot 10^{-10}$ m s^{-2}. With $H_0 = 73.04$ it is $1.1827\cdot 10^{-10}$ m s^{-2}.
- Two values of H_0 are used, which come from different measurements. H_0 is measured in kilometres per second per megaparsec.
- Sources: MM 2.54 and MM 2.56.

The baryonic surface density.

- Σ_b = 47 M☉ pc^{-2}. It is how much ordinary matter there is in the disc of the Galaxy, around the Sun, per square parsec.
- *Baryonic* means made of ordinary matter, the matter of atoms. M☉ is the mass of the Sun, used as a unit.
- Source: Park et al. 2026, after Bland-Hawthorn and Gerhard 2016.

The pairs of stars.

- They are 919 low-acceleration pairs (s > 5000 au, x < 1) and 21,471 control pairs.
- s is the *separation*, that is, the distance between the two stars. The au (*astronomical unit*) is the mean distance between the Earth and the Sun.
- x is the Newtonian acceleration of the pair divided by a_0. x < 1 means that this acceleration is below a_0.
- The *control* pairs are closer, with higher acceleration. They serve for comparison.
- Source: MM 2.56 (El-Badry, Rix and Heintz 2021).

The eccentricity law of the pairs.

- p(e) ∝ e^α, with α(s) of Hwang et al. 2022.
- The *eccentricity* e says how flattened an orbit is. 0 is a circle. Near 1 it is a very elongated ellipse.
- p(e) says how frequent each eccentricity is. α(s) is an exponent that changes with the separation s.
- As in MM 2.56.

The supplementary perihelion precessions.

- The *perihelion* is the point of a planet's orbit closest to the Sun. *Precession* is a slow rotation of the orbit. *Supplementary* means in addition to the expected one.
- Source: EPM2011 (Pitjeva and Pitjev 2013), with the INPOP10a uncertainties (Fienga and Minazzoli 2024).
- EPM2011 and INPOP10a are *ephemerides*: large tables that give the positions of the planets.

Light deflection.

- The Sun's gravity bends the light that passes close to it. The number γ (gamma) says how much it bends it. In general relativity $\gamma = 1$.
- Measurement: $\gamma - 1 = (2.1 \pm 2.3) \cdot 10^{-5}$.
- Source: Bertotti, Iess and Tortora 2003 (via Will 2014).

The planetary elements.

- They are the numbers that describe the orbits of the planets. Here: J2000, table 1. J2000 is the reference date.
- Source: the approximate positions of JPL, a NASA laboratory.

Weak lensing.

- The KiDS-1000 lensing rotation curves. They are in 4 mass bins, with 60 points and the full covariance.
- The *covariance* says how much the errors of different points go together.
- Source: Brouwer et al. 2021 (as in MM 2.44).

The rotation curves. SPARC, tables 1 and 2. Source: Lelli, McGaugh and Schombert 2016.

The environment field of the SPARC galaxies. log e_N,env, table 3. Source: Chae et al. 2021 (arXiv:2109.04745).

The wide binaries for the direction.

- They are 57,583 pairs beyond 5,000 au, and the catalogue beyond 300 au.
- Source: El-Badry, Rix and Heintz 2021 (VizieR J/MNRAS/506/2269). VizieR is an archive of astronomical catalogues.
- The catalogue was queried in August 2026.

The dwarf galaxies of the Local Group.

- The *Local Group* is the group of galaxies to which the Milky Way belongs.
- The data are: σ★, V, E(B−V), r_h, ellipticity, distance modulus.
- σ★ is the *velocity dispersion* of the stars: how different their velocities are from one another.
- V is the *magnitude* in the colour band V, that is, how bright the system appears.
- E(B−V) measures how much the dust in space, between us and the system, reddens the light.
- r_h is the radius that contains half of the light.
- The ellipticity says how flattened the system is.
- The distance modulus is a way of writing the distance.
- Source: McConnachie 2012, version of January 2021 (CADC, a Canadian archive of astronomical data).

The globular clusters.

- The data are: σ_0, mass and M/L_V (for L_V), r_h, R_GC.
- σ_0 is the velocity dispersion at the centre.
- M/L_V is the ratio between the mass and the light. L_V is the light in a colour band called V.
- R_GC is the distance from the Galactic centre.
- Source: the Baumgardt database, combined table of January 2026.

Dwarfs and clusters with metallicity spreads.

- The data are: σ_v, σ[Fe/H], r_½, M_V, distances, host.
- σ_v is the velocity dispersion.
- σ[Fe/H] is the metallicity spread, measured with iron.
- M_V is the *absolute magnitude*, a measure of luminosity.

- The *host* is the galaxy around which a system goes.
- Source: Local Volume Database (Pace 2025), version of 12 August 2026.

The papers and the larger data files are not included in the package. A list gives them, with their address online (see the final section on the work).

3. The QUMOND chain taken to the Sun

Here the chain of MM 2.56 is taken all the way to the Sun. The question is: which quadrupole does it predict?

The outcome, in the sentence written before the computation: the chain of MM 2.56, taken to the Sun with the same field equation, is not compatible with Cassini ($Q_2 = 3.22 \cdot 10^{-26}$ s^{-2}, z = 17.0).

The number z says by how many typical errors the prediction lies away from the measurement. z uses the Cassini error.

The computation can be done with different choices. There are three choices.

- **The value of H_0:** 67.4 or 73.04.
- **The external field:** V^2/R or the Gaia value.
- **The conversion to Newtonian.** The measured field already contains the boost. To get the Newtonian field, e_N, it has to be removed. To do this a hypothesis on the shape of the Galaxy is needed: *spherical*, that is, like a ball, or *with disc*.

H_0	external field	conversion to Newtonian	e_N	Q_2 (10^{-26} s^{-2})	z
67.4	V^2/R	spherical (**main case**)	1.351	**3.221**	**17.0**
67.4	V^2/R	with disc	1.369	3.254	17.2
67.4	Gaia	spherical	1.501	3.499	18.6
67.4	Gaia	with disc	1.517	3.526	18.7
73.04	V^2/R	spherical	1.210	3.328	17.6
73.04	V^2/R	with disc	1.228	3.367	17.8
73.04	Gaia	spherical	1.347	3.625	19.3
73.04	Gaia	with disc	1.362	3.658	19.4

Let us read the table. e_N is the Newtonian field of the Galaxy, divided by a_0. Q_2 is written in units of 10^{-26} s^{-2}.

- The **main case** is the first row: $H_0 = 67.4$, field V^2/R, spherical conversion. It gives e_N = 1.351, $Q_2 = 3.221$ and z = 17.0.
- With the disc: e_N = 1.369, $Q_2 = 3.254$, z = 17.2.
- With the Gaia field, spherical: 1.501, 3.499 and 18.6. With the disc: 1.517, 3.526 and 18.7.
- With $H_0 = 73.04$ and V^2/R, spherical: 1.210, 3.328 and 17.6. With the disc: 1.228, 3.367 and 17.8.
- With $H_0 = 73.04$ and Gaia, spherical: 1.347, 3.625 and 19.3. With the disc: 1.362, 3.658 and 19.4.
- In all eight cases z lies between 17.0 and 19.4.

The method. The computation uses the *phantom-density moment*.

- In QUMOND the extra gravity behaves as if there were an invisible mass in empty space. This mass that is not there is called the *phantom density*.
- The *moment* is a way of adding it up, with a weight that depends on the position, to obtain Q_2.
- The method reproduces Park's values exactly: 3.387 and $3.411 \cdot 10^{-26}$.
- The MM 2.56 solver gives the same Q_2 within $2.8 \cdot 10^{-5}$.

The boost at the Sun. The chain asks for a *radial* boost of about 1.45 at the Sun. Radial means in the direction of the radius. Park and colleagues, within QUMOND, allow at most 1.010 to 1.018.

Where the signal is generated.

- r_M is the *MOND radius* of the Sun. It is the distance at which the Sun's Newtonian acceleration drops to a_0.
- For the Sun r_M = 7,371 au.
- The table says which part of Q_2 is generated at each distance. And it says how many pairs there are there.

r/r_M	share of Q_2	low-band pairs	control pairs
< 0.30	8.2 %	0	21,461
0.30–0.95	61.2 %	0	10
0.95–1.93	24.2 %	799	0
> 1.93	6.4 %	120	0

The *low band* is that of the 919 low-acceleration pairs. Let us read the table row by row.

- Below 0.30 r_M, 8.2 % of Q_2 is generated. There are 0 low-band pairs there and 21,461 control pairs.
- Between 0.30 and 0.95 r_M, 61.2 % is generated. There are 0 low-band pairs and 10 control pairs.
- Between 0.95 and 1.93 r_M, 24.2 % is generated. There are 799 low-band pairs and 0 control pairs.
- Beyond 1.93 r_M, 6.4 % is generated. There are 120 low-band pairs and 0 control pairs.

In the scale s/r_M the low-band pairs sample the outer part of the region where the quadrupole is generated. In words: the far-apart pairs sit where the outer part of the signal is generated.

The precessions.

- The same Q_2 gives Saturn a perihelion drift of +4.5 mas/century. *Drift* means that the perihelion turns slowly.
- The mas is a thousandth of an arcsecond. It is a tiny angle.
- The EPM2011 ephemerides give instead -0.32 ± 0.47.

Light deflection. The *metric sector* gives $\gamma = 1$ by structure, as measured.

- The metric sector is the part of the theory that says how gravity curves space and time.
- *By structure* means that $\gamma = 1$ comes from the very form of the theory, without tuning any number.

4. The local reading

This is the central part of the record. Here the rules of the local reading are presented one by one, with their checks.

4.1 Where the anomaly lives

- In QUMOND the anomaly is a phantom density in the empty space around a mass. Around the Sun it forms a cloud thousands of au wide. This cloud generates the quadrupole excluded by Cassini.
- In the local reading, instead, the anomaly lives in the relation between two material bodies.
- The phantom cloud is a tool of the QUMOND equation, not an object.

An everyday example. On maps there are meridians and parallels. They serve to do calculations and to find one's way. But on the ground those lines are not there.

4.2 Who reads the external field: the body whose local MCD prevails

The rule. The local MCD of a mass is R_s/r.

- In a relation there are two bodies. The MCD of each is taken at the place of the other, that is, at the same distance r.

- The Schwarzschild radius is proportional to the mass. The ratio of the two MCDs is therefore the mass ratio.
- The body whose MCD prevails reads the external field for its whole system.

An example with numbers chosen by us. One body has a mass 10 times larger than another. At the same distance, its MCD is 10 times larger. It is the one that prevails.

The Solar System. The Sun's MCD at a planet exceeds the planet's MCD at the Sun. It exceeds it by as many times as the Sun's mass exceeds the planet's:

- 1,048 times for Jupiter;
- 3,499 times for Saturn;
- more for the other planets;
- about 10^{17} times for a comet.

Planets and comets read the Galaxy through the Sun. So the Sun and Saturn receive the same acceleration from the Galaxy.

- Without this rule, the two accelerations would differ by $6.7 \cdot 10^{-11}$ m s^{-2}.
- As a scale estimate: $\delta a/n^2 \approx 1{,}500$ km for Saturn's orbit. Here δa is that difference in acceleration. The letter n is Saturn's angular speed, that is, how fast it goes round its orbit. In words: the scale of the shift would be about 1,500 km.
- Radio tracking, instead, measures at the level of metres.

Galaxies. A galaxy prevails over its stars and its gas. The mass ratios go from 10^6 to 10^{11}. 10^6 is a million. Stars and gas read the environment field through the galaxy.

Wide binaries. In the pairs the mass ratio is at most 7.4. So each star reads the Galaxy itself.

The threshold. What does *prevailing* mean? Any threshold between 7.4 and 1,048 will do. No number has to be tuned. In words: 7.4 is the largest ratio between the stars of the binaries. 1,048 is the smallest between the Sun and the planets. Any threshold in between separates the two cases.

4.3 A dominant and compact body reads the relation itself

The rule. Sometimes a body prevails and is also *compact* with respect to the relation. That is, its radius, R_dom, is much smaller than the distance r. Then the relation is read with the MCD of the dominant body at its own place: its surface.

- There the reading is very high: for the Sun $x = 2.5 \cdot 10^{12}$. Remember: x is Newton's acceleration divided by a_0.
- So the relation is Newtonian. With x so large, the boost ν is practically 1.

An extended dominant body is, for example, a galaxy around its stars and its gas. It is spread along the relation. So the reading stays local.

The compactness threshold. R_dom/r is the ratio between the radius of the dominant body and the distance. Any threshold between two values will do.

- $7 \cdot 10^{-4}$: it is the case of Sgr A* and S2 at *pericentre*, the point of the orbit closest to the black hole. It is the least compact of the compact cases.
- $2.5 \cdot 10^{-2}$: it is the case of the Milky Way with a satellite at 100 kpc. It is the most compact of the extended cases.

What it gives.

- **Planets and comets on any orbit** are Newtonian. This also holds for the new comets of the *Oort spike*. They are comets that have never passed close to the Sun. The *Oort cloud* is a large cloud of comets at the edges of the Solar System.
- **S2:** no anomalous boost. So general relativity remains, as measured.
- **Wide binaries:** no star prevails. So the boost remains.
- **Galaxies:** the radial acceleration relation remains. This also holds for the outer gas and for the satellites.

4.4 A relation without a dominant body: the inertial state

The rule. If no body prevails, the reading of the relation is its *time average* along the orbit:

$x_eff = \langle x \rangle = x(a)/\sqrt{(1 - e^2)}$, with $x = GM/(r^2 a_0)$.

- x is Newton's acceleration, GM/r^2, divided by a_0. G is the gravitational constant, M the mass, r the distance.
- a is the *semi-major axis*: half of the longest diameter of the ellipse of the orbit.
- e is the eccentricity of the orbit.
- In words: the effective reading, x_eff, is the x computed at the semi-major axis. Then it is divided by the square root of (1 minus e squared).

An everyday example. The temperature changes during the day. But to say what a day was like, the mean temperature is often used. Here the same is done with x, along the whole orbit.

- The rule applies to relations in which no body prevails, such as wide binaries.
- On DR3 the orbit average fits the data better than the *instantaneous* reading, that is, the one taken point by point (section 5).
- On circular orbits x_eff = x. So the radial acceleration relation of disc galaxies is unchanged.

The DR3 gain sits in the fast pairs. On DR3 the gain over Newton comes entirely from the fast pairs.

- The pairs are divided into *u classes*. The number u measures how fast a pair is.
- The third with the highest u gives +10.6 of a gain of about +11. It is an average over four series of simulations.

A reading that falls. Another reading can be thought of.

- In the fast phases of the orbit the *centrifugal* force prevails over the gravity read. The centrifugal force is the outward push that one feels on a bend, for example in a car.
- This reading would make the fast phases Newtonian. It would remove the boost exactly there.
- The data exclude it. The registered criterion gives −10.4 (s 3.2), instead of a positive value. Here −10.4 is the average over the four series of simulations, and s is the standard deviation: how much the result changes from one series to another. This reading falls.
- The u classes alone do not say in which phase of the orbit a pair is.

4.5 How the interior and exterior fields add up

The boost of a relation is $\nu(x_eff + e_N)$, with:

- $a_0 = cH_0/6$;
- e_N = 1.351. It is the field of the Galaxy at the Sun, in Newtonian form, divided by a_0. It is obtained from $V^2/R = 1.966\ a_0$ through the law ν.

In words: the interior reading of the pair, x_eff, is added to the external field, e_N. Then the law ν is applied.

The forms compared. There are several ways of adding interior and exterior. The record registered four of them:

- the *vector sum*: they are added like arrows, taking the directions into account;
- the *quadrature*: the squares are added and then the square root is taken;
- the *sum of magnitudes*: the values are added, without looking at the directions;
- the *relation without exterior*: the external field does not enter.

An example with numbers chosen by us. Two pushes are worth 3 and 4, and they are at a right angle. Added as arrows they give 5. Added as magnitudes they give 7.

In the instantaneous form, the sum of magnitudes gives the pairs a boost close to the chain's: +2 %. The other forms give 28–203 % more.

With the inertial state, that is, with the average along the orbit, the per-pair mean is $\langle \eta \rangle = 1.27$. It is an excess 15 % below the chain's.

- This estimate takes the observed separation as the semi-major axis.

- In the full forward comparison on DR3, with whole orbits, the reading fits the data better than the chain (section 5).

The external field read by matter is the baryonic one, that is, the field of ordinary matter.

- One can try with the *total field*. The total field also includes the dark-matter mirror: it is written e_N ν(e_N).
- With the total field the SPARC curves worsen by $\Delta\chi^2 = +2{,}757$ and the dwarfs worsen too. $\Delta\chi^2$ is a difference of χ^2: a positive number means worse.
- In the wide binaries, instead, the full forward model on DR3 does not separate the two fields. The test is this. The total field is put in place of e_N inside ν: it is the reading under test.
- The likelihood changes by −1.4 against the mean of the 2.60 reading. The registered thresholds are ±6.98: the change lies within them.
- Against the QUMOND chain the difference is +2.5.

No significant dependence on the mass ratio, below the prevalence threshold.

- On DR3 the sample effect seen in the close pairs is removed first.
- Then the *unequal* pairs ($q < 0.5$) are compared with similar ones. Here q is the ratio between the masses of the two stars.
- The difference is -0.03 ± 0.05 dex.

The value of the Galactic field.

- The Gaia field can be used, about 8 % stronger than V^2/R.
- Then the mean boost of the 919 pairs goes from 1.2745 to 1.257.
- The excess is 6–7 % smaller.

The screening readings. *To screen* means to shield, as an umbrella shields from the rain. Four readings of this kind were also registered:

- the charge reserve;
- the crossing filter;
- the 30 pc cell;
- the history factor.

These readings silence Saturn, that is, they remove the anomaly on Saturn. But they make the binaries Newtonian too. This is the declared alternative of MM 2.56.

4.6 Direction: a tensorial alignment

Here the direction of the pair with respect to the Galactic centre comes into play.

The Galactic MCD difference between the two stars. The star nearer the Galactic centre reads a higher Galactic MCD.

- Along the line joining the two stars, the difference is $2\, g_e\, s \cos\theta / c^2$.
- Here g_e is the external field and s the separation. θ (theta) is the angle between the pair and the direction of the centre. c is the speed of light.
- The pair's own MCD is $2GM/(s\,c^2)$.
- Their ratio, $q = (g_e/a_0)/x$, sets the weight of the direction. Careful: this q is not the mass ratio of section 4.5. It is another number, with the same letter.
- The *median* of q is 3.2 for the 919 wide pairs. It is 0.018 for the close pairs. The median is the value that sits in the middle: half of the pairs have a smaller q, half a larger one.

The reading is a tensorial alignment. A *tensor* is a mathematical object that takes two directions into account together.

- Here there are two directions: $\hat{s}$, the line of the pair, and $\hat{e}$, the direction of the centre.
- The line of the pair *contracts* with the direction of the centre: $\hat{s}_i\hat{s}_j\hat{e}_i\hat{e}_j = \cos^2\theta$. To contract means to combine the two directions into a single number.
- In words: the result is the square of the cosine of the angle between the two directions. It is 1 when the pair points towards the centre. It is 0 when the pair lies crosswise.

More MCD read means more speed:

$v^2 = v_0^2 (1 + w\ q \cos^2\theta)$.

In words: the square of the speed, v^2, is the starting one, v_0^2, multiplied by (1 plus w times q times $\cos^2\theta$). The boost is largest when the pair is aligned with the Galactic centre.

The amount w is an observational input. That is, it is measured from the data, not derived from the theory.

- The data of August 2026 give $w \approx 0.007$. The interval at 2σ goes from 0.001 to 0.014. This value comes from a 3D statistic, that is, one made in three dimensions.
- From the quadrupole on the sky comes instead $w = 0.024 \pm 0.022$.
- With $w \approx 0.007$ the mean boost of the wide pairs changes by +0.002 dex. So section 4.5 is not affected by it.
- The simulations registered for DR4 do not include this term.

The computation redone with the program for DR4.

- The program for DR4 was run on DR3. It subtracted the level of the close pairs. And it used the full covariance (section 5.5).
- It gives $w = 0.005 \pm 0.010$ in the symmetric form, and -0.003 ± 0.009 in the dominant form.
- These values are compatible with zero. The upper limits are 0.024 and 0.014.
- The August values, written above, contained the sample level of the close pairs.

Two words for the sky: dipole and quadrupole. On the sky one looks at an angle, φ (phi), between the pair and the direction of the Galactic centre.

- A *cos φ* term makes a difference between the side towards the centre and the opposite side. It is called a *dipole*.
- A *cos 2φ* term is the same on the two opposite sides, and different crosswise. It is called a *quadrupole*. It is like the tide, which raises the water on two opposite sides of the Earth.

The form is symmetric. Imagine the companion behind the star, on the far side from the centre. The star in front still moves with the anomaly, because of the companion. The boost is there on both sides. So on the sky there is only the cos 2φ term, and the dipole is zero.

The alternative form is registered too. It is read by the dominant star, as in section 4.3.

- Here the boost is there only when the *secondary* star lies between the *primary* and the centre. The primary is the main star of the pair, the secondary its companion.
- This would give a dipole (cos φ) about 1.9 times the quadrupole.

The data of August 2026. They are 9,378 pairs between 5,000 and 50,000 au, in the region $w > 0.5$.

- The dipole is $+0.0027 \pm 0.0046$ dex. The quadrupole is $+0.0050 \pm 0.0046$ dex.
- In the close pairs (300–2,000 au) the quadrupole is $+0.0041 \pm 0.0018$ dex, where the prediction is zero. The dipole is $+0.0001 \pm 0.0018$ dex.
- The *Galactic north pole* is used as a control direction. It is the direction perpendicular to the disc of the Milky Way. Nothing is seen there.

The dipole is clean. Gaia observes the sky following a fixed path, its *scanning law*. Neither this law nor the selection of the pairs knows which star is the brighter with respect to the Galactic centre.

4.7 Light reads the curvature

The rule. Light reads the *curvature* of the MCD, with the same law as the stars, without extra mass. The curvature says how much a field changes from one point to another, not only how much it is worth.

Light does not read the external field. A *uniform* field, that is, the same at every point, has no curvature. Matter instead reads relations, and there the external field enters.

An everyday example. In a lift in free fall, you and the objects fall together. From inside, a uniform field is not noticed. Only the differences from one point to another are noticed.

KiDS-1000 lensing (Brouwer et al. 2021, 60 points, full covariance).

- $\chi^2 = 219.9$ with $a_0 = cH_0/6$, against 1,687.8 for Newton.
- The program reproduces the MM 2.44 values: 236.61 and 1,687.81.

A reading that falls. One can try to add the environment field to light, as for matter.

- Then the signal between 1 and 2.6 Mpc is lowered by a factor of 9 to 17. The Mpc (megaparsec) is a million parsecs.
- This addition is excluded for all the values tried, from 0.003 to 0.1 a_0 ($\Delta\chi^2 \geq 552$). The sign ≥ means greater than or equal to.

As in MM 2.44, no model fits the data fully with the covariance provided: $\chi^2/60 \approx 3.7$. In words: the χ^2 per point is about 3.7. A full agreement would give a value close to 1.

Denser environments and central galaxies.

- KiDS-1000 compares all lenses with isolated lenses (Brouwer et al. 2021, Fig-A4 and Fig-9).
- The ratio between the two signals goes from 2.0 to 3.0 at the lowest g_bar. It returns to 1 above $4 \cdot 10^{-14}$ m s^{-2}. g_bar is the acceleration given by ordinary matter alone.
- The excess grows with R, the distance from the centre of the galaxy. The slope is -0.90 ± 0.14 in g_bar. In words: the smaller g_bar is, that is, the farther one goes, the more the excess grows.
- Isolated lenses have no brighter galaxies within 3 Mpc. So they are not satellites of larger galaxies.
- Among isolated lenses, red and blue galaxies differ by a nearly constant ratio. The same holds for galaxies with high and low *Sérsic index*. This index describes how the light of a galaxy falls off, from the centre towards the edge. The slopes are -0.28 ± 0.10 and $+0.02 \pm 0.09$.
- A growth as R^2 (slope −1) is disfavoured: $\Delta\chi^2$ +16 and +21.
- The errors treat the two samples as independent. The covariance due to shared *sources* is not provided. The sources are the distant galaxies whose images are deformed.

The satellites.

- The growing excess of the full sample is compatible with its satellites. For a satellite, the host galaxy enters the signal only beyond the distance between the two.
- The comparison between red and blue does not prove it. The density difference of their environments, in fact, is not measured.
- A distortion that grows as R^2 does not appear among isolated lenses.

4.8 Galaxies in their environment

The rule. The environment field enters in full and adds as a magnitude, as in the binaries.

78 SPARC galaxies, with 1,419 points. For each galaxy its own environment field is used, taken from Chae et al. 2021. Three ways are tried:

environment field	$\Sigma\chi^2$	$\Delta\chi^2$
in full, added as a magnitude	**12,490.8**	**0**
diluted by the ratio of the galaxy's mean MCD to a star's	13,341.2	+850.4
none	13,447.5	+956.7

$\Sigma\chi^2$ is the sum of the χ^2 of all the galaxies. The symbol Σ (capital sigma) means sum. $\Delta\chi^2$ is the difference from the best way.

- **Field in full, added as a magnitude:** $\Sigma\chi^2 = 12{,}490.8$. It is the best: $\Delta\chi^2 = 0$.
- **Diluted field**, that is, reduced by the ratio of the galaxy's mean MCD to a star's: $\Sigma\chi^2 = 13{,}341.2$, that is, +850.4.
- **No environment field:** $\Sigma\chi^2 = 13{,}447.5$, that is, +956.7.

The size of the effect. The environment field has a median of $1.9 \cdot 10^{-3}$ a_0. At x = 0.01 it lowers g by 9 %. Here g is the acceleration.

Concentration. Much of the advantage comes from a few galaxies.

- Five *declining* curves, that is, with the speed that drops towards the edge, give 865 of the 957 points of $\Delta\chi^2$.
- Without them the full field still leads by 92.
- Of the remaining 73 galaxies, 38 improve and 35 worsen.

Fit quality. Distances and inclinations are fixed at the SPARC values. So the χ^2 per point is about 9. The *inclination* is how much the disc is tilted with respect to us.

4.9 Systems without rotation: dwarf spheroidals and globular clusters

The reading. In these systems the stars do not rotate in order. They move in all directions.

- A star reads the relation with the rest of the system along its line through the centre.
- The increase of MCD is set by the local state: ν(x_h + e_N).
- Here x_h is the x of the system. e_N is the external field. Both are divided by a_0.

The check.

- There are two samples. The first: 35 Milky-Way satellites, between 20 and 300 kpc (McConnachie 2012, January 2021 version). The second: 25 globular clusters beyond 20 kpc (Baumgardt database, January 2026).
- The method uses the *Wolf estimator* (Wolf et al. 2010). It is a formula that obtains the mass from the velocity dispersion and the radius.
- The visible mass is M_b = Υ★ L_V. Υ★ (upsilon star) is the stellar *mass-to-light ratio*: how much mass there is for each unit of light. It is 2 for the dwarfs and 1.7 for the clusters. L_V is the light of the system.
- The Milky Way counts as $6\cdot10^{10} M_\odot$ baryons, that is, sixty billion solar masses.
- Nothing is fitted: no number is tuned on the data.

sample	Newton	ν(x_h + e_N)	ν(x_h), no external field	boost on one axis only
dwarfs (35)	+1.00 (χ^2 1,629)	+0.54 (χ^2 598)	+0.31 (χ^2 200)	+0.68 (χ^2 939)
clusters (25)	+0.05 (χ^2 27)	−0.16 (χ^2 163)	−0.29 (χ^2 255)	−0.04 (χ^2 64)

The entries are the median of log10(σ_obs/σ_pred), in dex.

- σ_obs is the observed velocity dispersion. σ_pred is the predicted one.
- log10 is the base-ten logarithm.
- A value of 0 means perfect agreement. A positive value means that the stars move faster than predicted.

Let us read the table, column by column.

- **Newton:** dwarfs +1.00 (χ^2 1,629); clusters +0.05 (χ^2 27).
- **ν(x_h + e_N)**, the reading of the record: dwarfs +0.54 (χ^2 598); clusters −0.16 (χ^2 163).
- **ν(x_h)**, without external field: dwarfs +0.31 (χ^2 200); clusters −0.29 (χ^2 255).
- **Boost on one axis only:** dwarfs +0.68 (χ^2 939); clusters −0.04 (χ^2 64).

The classical dwarfs, with ν(x_h + e_N):

- Fornax −0.08, Leo I −0.04, Leo II +0.05, Sculptor +0.11, Carina +0.15;
- Draco, Ursa Minor and Sextans: from +0.40 to +0.42.

The ultra-faint dwarfs: from +0.4 to +1.3 dex.

- The *residual*, that is, the gap between measurement and prediction, grows with the weight of the Milky-Way field.
- The Spearman coefficient is ρ = +0.73. It is a number between −1 and +1. It says how much two quantities grow together.

The globular clusters are Newtonian even at 100–150 kpc: Palomar 14 +0.06, NGC 2419 +0.03, Crater −0.04, AM 1

−0.13.

Same gravitational state. x_h + e_N is taken in the range common to the two samples, between 0.019 and 0.12.

- 19 dwarfs sit +0.995 dex above Newton (+0.690 with ν(x_h + e_N)).
- 11 clusters sit +0.022 dex above Newton (−0.387 with ν(x_h + e_N)).

The mass-to-light ratio. Y★ is changed by a factor 10^(±0.15), up and down. The conclusions do not change:

- dwarfs with ν(x_h + e_N): from +0.62 to +0.47;
- clusters: Newtonian in all three cases.

Without the external field the dwarfs fit better: median +0.31, χ^2 200 against 598. But they stay above the prediction.

Who carries the mirror: the systems with stars of different generations.

- The stars of a dwarf were born at different times. Those of a cluster were born nearly together.
- An everyday example. A dwarf is like a family with grandparents, parents and children. A cluster is like a school class, where everyone is almost the same age.
- In the reading, this distance in time between the stars gives the density, and so the MCD. The mirror depends on this MCD.
- The data show the association. The metallicity spread is also one of the criteria astronomers use to separate galaxies from clusters.
- In the Local Volume Database all 38 systems with a measured metallicity spread carry a mirror: B = M_dyn/M_b > 3. In words: the dynamical mass is more than three times the visible mass.
- Two of them are catalogued as clusters: Balbinot 1 and DELVE 4.
- Clusters with nearly zero spread are Newtonian. For them the spread has only limits, from 0.05 to 0.14 dex.
- Within galaxies the amount of mirror does not follow the spread (ρ = +0.01).

How much mirror: a common profile set by the radius. The dwarf is an element on its own. It has an autonomous mirror, whose extent is given by its radius.

- **The formula.** For 81 dwarfs (satellites of the Milky Way and of M31, and isolated dwarfs): log M_dyn = 7.064 + 1.53 (log r_½ − 2.5). M31 is the Andromeda galaxy. In words: the dynamical mass is $3\cdot10^5$ M⊙ at 30 pc, $1.1\cdot10^7$ at 300 pc and $6.8\cdot10^7$ at 1 kpc.
- **The exponent** is 1.54 ± 0.11. This value uses an error model in which the radius error enters M_dyn and r_½ together. Without the gas-rich systems (measured HI) it is 1.40 ± 0.11. HI is neutral hydrogen, a gas.
- **The excluded exponents.** In both cases the exponents 2 and 1 are excluded, and 1.5 is compatible. Exponent 2 means constant acceleration: the mass grows as r^2. Exponent 1 means constant velocity: the mass grows as r.
- **Visible mass and metallicity.** The amount of mirror grows much less than the visible mass. And it does not follow the metallicity spread. At fixed radius a weak dependence on the visible mass remains (r = +0.39 at fixed tide). Here r is a correlation coefficient, not the radius. The *tide* is the force with which the Milky Way tends to stretch the system.
- **Satellites and isolated dwarfs.** Satellites and isolated dwarfs lie on the same profile. Without the gas-rich systems, however, only three isolated dwarfs remain: Tucana, Cetus and Andromeda XVIII. The difference is −0.03 dex, with an interval from −0.50 to +0.13.
- **The intrinsic scatter.** Around the profile an intrinsic scatter of 0.48 ± 0.10 dex remains.
- Part of it is associated with the Milky-Way tide: there is less mirror where the tide weighs more (r = −0.41, p = 0.02, at fixed visible mass). The number p is the probability of finding such a strong link by pure chance.
- Part may come from binaries and from *non-equilibrium* among the ultra-faints. Non-equilibrium means that the system has not yet settled.
- **The common acceleration.** At r_½ the dwarfs share an acceleration of about 0.17 a_0. It is of the order of the Milky-Way field at those distances, about 0.1 a_0. But the dwarfs do not follow it one by one (slope 0.32).
- **The number 3/2.** The measured exponent is compatible with 3/2, halfway between 1 and 2: it is their arithmetic mean. The profile M ∝ r^(3/2) is the geometric mean of the profiles M ∝ r and M ∝ r^2. The geometric mean of two numbers is the square root of their product. This is an observation, not a derivation.

An example with numbers chosen by us: r = 4. Then r is 4, r^2 is 16 and r^(3/2) is 8. And 8 is exactly the square root of 4 times 16, that is, of 64.

The limits of the data.

- The dispersions of the ultra-faints can be inflated by binaries and tides.
- Y★ is fixed.
- The σ_0 values of the clusters come from models.

4.10 What it gives: the summary

Here are all the results of the local reading, system by system.

- **Saturn (Cassini):** $Q_2 \approx 10^{-30}\ s^{-2}$, from the Galactic tide only. The measurement is $(1.6 \pm 1.8)\cdot 10^{-27}$.
- **Planets, comets on any orbit, S2:** Newtonian (dominant compact body).
- **919 wide binaries**, between 0.95 and 3.7 r_M, with eccentricity e ~ 0.7: $\langle\eta\rangle = 1.27$. The QUMOND chain gives 1.32, Newton 1. The sign ~ means about.
- **Wide binaries, direction:** on DR3, once the close-pair level is subtracted, compatible with zero (w < 0.024).
- **Disc galaxies:** radial acceleration relation unchanged; environment field in full (SPARC).
- **Eccentric orbits inside a galaxy:** +6 % on the mean reading, for r_a/r_p = 5.7, with no point mass at the centre. r_a and r_p are the largest and the smallest distance from the centre.
- **Light around galaxies:** curvature, no external field: χ^2 219.9 on 60 KiDS points.
- **Light around galaxies in denser environments:** more signal at large radii (all against isolated). No growth as R^2 among central galaxies.
- **Classical dwarf spheroidals:** within ±0.15 dex.
- **Ultra-faint dwarfs:** 2.5 to 20 times above the prediction.
- **Globular clusters beyond 20 kpc:** Newtonian (Palomar 14 +0.06 dex, NGC 2419 +0.03).
- **Dwarfs, who carries the mirror:** all 38 systems with a metallicity spread.
- **Dwarfs, how much mirror:** M_dyn ∝ r_½^1.54±0.11 (1.40 without gas-rich systems), intrinsic scatter 0.48 dex.

5. What is fixed in advance for the new data

This section describes the tests registered for Gaia DR4, the next Gaia data release, planned for 2 December 2026. They are fixed before the data arrive. There are two tests. The first concerns the boost (sections 5.1-5.4). The second concerns the direction (section 5.5).

5.1 The test of the boost

The law in the simulations. The simulations compute the orbits of the pairs on a computer.

- Each relative orbit is *Keplerian*, that is, an ellipse like those of the planets, with G_eff = ν(x_eff + e_N).
- G_eff is the *effective* gravitational constant, measured in units of Newton's constant G. So G_eff = ν means: the gravity of the pair is Newton's multiplied by the boost ν.
- The same programs as in MM 2.56 are used. Only the force law and the file paths change.

The library. It contains 200,000 orbits, computed with seed 18 in 337 s (seconds).

- In a Keplerian orbit the energy stays constant. In the computation its relative change, |dE/E|, is $4.5\cdot 10^{-13}$ at the 99th percentile. It means that in 99 per cent of the orbits the change is smaller than that.
- Each orbit is Keplerian.
- No pair starts unbound, that is, with the stars free to move apart for ever.

The check of the copies. The programs are copies of those of MM 2.56. With the six hypotheses of MM 2.56, the evaluation reproduces the published *dry run* exactly: D = +7.64, subsets +7.56 / +4.50 / +2.07. A dry run is a rehearsal on data already seen, to check that everything works.

The decision rule. D_L = L(local reading) − L(Newton).

- L is the log-likelihood. In words: D_L says by how much the local reading does better than Newton.
- The simulations are done in several *replicas*, that is, several times. D_L,k is the same quantity, redone with one replica removed. The *subsets* are the values obtained in this way.
- **Local reading favoured:** D_L ≥ +5 and every D_L,k ≥ +2. The rule asks that the result stays at least +2 even when any one replica is removed.
- **Newton favoured:** D_L ≤ −5 and every D_L,k ≤ −2. The sign ≤ means less than or equal to.
- **Otherwise:** inconclusive, that is, the test does not decide.
- **Not valid:** if a validity condition of MM 2.56 is not met.

Informative numbers only, which do not decide:

- F_L = L(local reading) − L(chain): by how much the local reading does better than the chain;
- the slope statistic S, over the two halves of the low band.

5.2 The dry run on DR3

Here the data are already seen: the grade is B*. Grade B* marks a result obtained on data that were already known. The registered simulation series, called R0, is used.

hypothesis	ΔlogL from the best	deviance control / low
local reading	**0**	29.9 / 21.9
MOND simple	−6.10	43.3 / 20.7
MOND standard	−6.93	27.7 / 38.0
QUMOND chain, $H_0 = 67.4$	−8.46	37.6 / 31.1
QUMOND chain, $H_0 = 73.04$	−10.28	44.8 / 27.6
series, $a_0 = 1.2 \cdot 10^{-10}$	−14.82	48.6 / 32.9
Newton	−16.10	36.8 / 47.2

Let us read the table.

- ΔlogL is the difference in log-likelihood from the best hypothesis. It is 0 for the best. The others have negative numbers.
- The *deviance* measures how far a hypothesis is from the data, like χ^2. It is given for the control band and for the low band.

Here are the seven hypotheses, from the best to the worst.

- **Local reading:** 0, the best. Deviance 29.9 in the control band and 21.9 in the low band.
- **MOND simple:** −6.10 (43.3 / 20.7).
- **MOND standard:** −6.93 (27.7 / 38.0). MOND simple and MOND standard are two ways of writing the transition between Newton and MOND.
- **QUMOND chain with $H_0 = 67.4$:** −8.46 (37.6 / 31.1).
- **QUMOND chain with $H_0 = 73.04$:** −10.28 (44.8 / 27.6).
- **Series, with $a_0 = 1.2 \cdot 10^{-10}$:** −14.82 (48.6 / 32.9). Careful: here *series* is the name of a hypothesis. It is the law ν with $a_0 = 1.2 \cdot 10^{-10}$, not a series of simulations.
- **Newton:** −16.10 (36.8 / 47.2).

The numbers of the rule:

- D_L = +16.10, with subsets +16.92 / +16.00 / +6.46. The rule would give "local reading favoured over Newton".
- F_L = +8.46, with subsets +9.35 / +11.50 / +4.39.

The statistic S. The low band is divided into two halves: the closer pairs and the wider ones. In each half the median u is looked at.

- For the reading: 0.331 and 0.358.
- In the data: 0.342 and 0.353.
- For the chain: 0.339 and 0.358.

The instantaneous form is registered too. It is the same composition, but without the inertial state. Its working name is K5c. On DR3 it gives D_K = +5.31 (subsets +2.07 / +9.04 / +9.43) and F = −2.33.

5.3 Simulation noise on DR3

The control band. In the close pairs the mean boost of the local reading is just 0.07 %. Yet their deviance changes by tens of points, when only the seed is changed.

- In the simulations there are also hidden companion stars and *twin* pairs, that is, pairs of two nearly identical stars.
- At zero companions, the twin fraction enters only the seed string.
- There the deviance of the local reading moves from 240.1 to 213.7, with the same physics. That of K5c moves from 313.6 to 222.4.
- The MM 2.56 simulations, which share the seed in this case, are identical.

Over the grid. The simulations are done on a *grid* of 30 points, that is, 30 combinations of starting values.

- Over these 30 points the difference local − Newton in the control band is looked at.
- It has mean −7.7 and spread 20.9.
- There are 15 positive and 15 negative values.

What it means.

- The differences between hypotheses in the control band are simulation noise. They are not a boost of the close pairs.
- The DR3 differences, D_L and F_L, carry a seed and library noise. The D of MM 2.56 carries it too.
- The subset rule does not measure this noise.

The measurement of the noise. The protocol and the programs were registered beforehand. The new series are these:

- R1 and R2: new seeds, with the same libraries;
- B1: new libraries for Newton (seed 21) and for the local reading (seed 22).

The checks:

- the program reproduces a registered simulation exactly. On R0 it gives D = +7.636 and D_L = +16.095;
- all 80 runs ended without errors.

series	D = chain − Newton	D_L = local − Newton	F_L = local − chain
R0 (registered)	+7.64	+16.10	+8.46
R1	+12.11	+10.12	−1.99
R2	−0.85	+8.20	+9.04
B1	—	+9.64	—
mean	**+6.30**	**+11.01**	**+5.17**
standard deviation	**6.58**	**3.49**	**6.21**

Let us read the table. D is chain − Newton. D_L is local − Newton. F_L is local − chain.

- **R0**, the registered series: +7.64, +16.10 and +8.46.
- **R1:** +12.11, +10.12 and −1.99.
- **R2:** −0.85, +8.20 and +9.04.
- **B1:** only D_L, which is +9.64.
- **Mean:** +6.30, +11.01 and +5.17.
- **Standard deviation:** 6.58, 3.49 and 6.21. The standard deviation says how far the values lie, on average, from their mean.

Against the criterion written beforehand. The criterion, written beforehand, was a standard deviation above 2.5. Against this criterion, the DR3 differences are not stable with respect to the seed, with the registered rule (three replicas

in one series).

- R0, R1 and R2 use the registered libraries. B1 adds new ones. So D_L has four values, while D and F_L have three.
- The local reading stays above +5 over Newton in all four series.
- The chain over Newton is the quantity of MM 2.56. With another series of seeds it would have given "inconclusive".
- The local reading and the chain are not separated in a stable way on DR3.

5.4 A rule over several series

For DR4 a new rule is registered, which uses several series of simulations.

The series.

- For the local reading and for Newton: seven series. They are R0–R4, with the registered libraries, and B1–B2, with new libraries.
- For the chain: five series.

The rule.

- Local reading favoured: the mean $\bar{D}_L \geq +5$ and $D_L \geq +2$ in every series. The small bar above the D marks the mean.
- Newton favoured: $\bar{D}_L \leq -5$ and $D_L \leq -2$ in every series.
- Otherwise: inconclusive.

The reporting. Both rules are reported: that of the single series R0 with its subsets, and the one over several series. The outcome is read with the rule over several series. The MM 2.56 rule for the chain stays as it was registered there.

The dry run on DR3, with the series that already exist:

- $\bar{D}_L = +11.0$ (minimum +8.2): "local reading favoured";
- $\bar{D} = +6.3$ (minimum −0.85) and $\bar{F}_L = +5.2$ (minimum −2.0): inconclusive.

The cost at DR4: 190 simulations, about five hours of computing.

The programs are registered.

- The first runs the 190 simulations. It stops with an error if a simulation fails or is missing.
- The second applies the rule, on every series. First it checks the MM 2.56 validity conditions: on the data, on the libraries, on the grid and on the fit.
- On DR3 all the conditions hold.

The checks on DR3.

- The copied forward program reproduces a registered MM 2.56 simulation exactly.
- The evaluation reproduces the numbers of the noise test: mean +11.01, minimum +8.20, "local reading favoured".
- The two new libraries are already generated: Newton with seed 23, the local reading with seed 24.

5.5 The test of direction: the anisotropy

The model. $v^2 = v_0^2 (1 + w\, q \cos^2\theta)$, as in section 4.6.

- The number w is measured in each of three wide bands: 5,000–10,000, 10,000–20,000 and 20,000–50,000 au.
- The projection on the sky uses the orbits of the MM 2.56 forward model.

Consistency. The three values must agree: χ^2 with 2 degrees of freedom, $p > 0.05$. The *degrees of freedom* count the independent comparisons. Moreover the common w must be positive.

The sample effects. In the 300–2,000 au band the reading predicts zero. The level measured there is subtracted, as a

constant.

An everyday example: the tare. Before weighing the flour, you weigh the empty bowl. Then you take away its weight.

The dipole is the primary statistic to tell the two forms apart.

- In the form read by the dominant star: positive, about 1.9 times the quadrupole.
- In the symmetric form: zero.
- The close-pair dipole must be zero. The Galactic north pole is the control direction.

Addendum 2. The reading predicts the symmetric form: a zero dipole in every band, with all the anisotropy in the quadrupole.

- A positive dipole of about 1.9 times the quadrupole would favour the alternative.
- On DR3 the dipole is +0.0027 ± 0.0046 dex.

The code was registered before the DR4 release.

- It reads the files written by the MM 2.56 data conversion.
- The dry run on DR3 gives S_ORB = 0.166 dex, as in August.
- The close pairs show B2 = +0.0037 ± 0.0015 and B1 = 0.0000 ± 0.0015. S_ORB, B2 and B1 are quantities computed by the program. Here B1 and B2 are not the series of section 5.3.
- The outcome of the dry run: w compatible with zero in both forms (limits 0.024 and 0.014).

5.6 What will be done when the new data come out

On 2 December 2026 the data are prepared with the steps of MM 2.56. Then these are done, in order:

- the 60 simulations of MM 2.56;
- the 10 + 10 simulations of the two local forms;
- the evaluation of the local reading and of the instantaneous form;
- the 190 simulations and the rule over several series (section 5.4);
- the anisotropy test (section 5.5).

The exact commands are in the final section, the one on the work.

6. Predictions the data will read

These are the predictions of the record. The data will say whether they hold.

Gaia DR4.

- The boost of wide binaries, and its rise with separation.
- The anisotropy. It must be positive, and only in wide pairs. It must grow as $q \propto s^2/M$. In words: it grows with the square of the separation, and it decreases if the pair is heavier.
- The same w in every band, and no change of sign. Full QUMOND, instead, changes sign near 12,000 au.
- A zero dipole (symmetric form). A positive dipole would favour the reading by the dominant star.
- No significant dependence of the boost on the mass ratio, below the prevalence threshold. This check will be made on many more unequal pairs.

Systems without rotation.

- Distant globular clusters Newtonian.
- New dwarfs with stars of different generations, on the same profile: M_dyn ∝ r_½^1.5.
- New single-generation clusters Newtonian.

Light companions.

- Think of a companion much lighter than its star, at a wide separation. For example a *brown dwarf*, that is, a failed star, too small to light up. Or a planet at 10^4 au, that is, at ten thousand au.
- This companion is read by the star itself, and it is Newtonian.

- Pairs with comparable masses are boosted.
- Between the mass ratios 1/1,048 and 1/7.4 the threshold will be set by the data.

Comets and the Oort cloud: Newtonian, on any orbit.

Weak lensing: no external-field effect around galaxies, in any environment. The *dynamics* of stars and binaries, that is, their motion, does show it instead.

7. What this record does not contain

This section says what the record does not claim and does not do.

- **A covariant field theory of the local reading.** A *covariant* theory has the same form for all observers, like general relativity. Here the local reading is a law of motion for *bound relations*, that is, for bodies that go round each other. To this a reading of light through the curvature is added.
- **The direction term inside the DR4 simulations of the boost.** Its mean effect is +0.002 dex.
- **A derivation of the common profile of the dwarfs.** Its exponent and its normalisation are measured, not obtained from the theory. The *normalisation* is the level of the profile: it says how large the mass is at a given radius.
- **A change to the MM 2.56 protocol.** The record changes nothing in that protocol.

How the work was done

This section gathers the procedure: who did the computations, the checks, the registrations, the files.

Who did the work. The author is not a professional physicist. The computations were done with AI assistants (Claude, with checks by Codex). No human referee reviewed them. A *referee* is an expert who checks a work before a journal publishes it.

The licences. Texts and materials: CC BY 4.0. Code: MIT. A licence says how other people can use the work.

The working names. In the working records this deposit is called 2.58. The reference texts, in Italian, are in the folder documenti/. The archived text in English, the README, summarises them.

The files cited in the sections.

- **Section 2, the data:** documenti/INVENTARIO_LIVELLO0_v0.md and _v1_AGGIUNTA.md; dati/FONTI.md.
- dati/FONTI.md lists with URLs and SHA-256 the papers and the larger data files, which are not included.
- The MM 2.56 sample of pairs is dati/campione_F.csv.
- The JPL approximate positions are in dati/approx_pos.html.
- The Local Volume Database file is comb_all.csv.
- **Section 3, the chain taken to the Sun:** documenti/PROTOCOLLO_LIVELLO1.md and NOTA_ESITO_LIVELLO1.md.
- **Section 4, the local reading:** the structure is in documenti/STRUTTURA_LETTURA_LOCALE.md, the numbers in documenti/CONTROLLI_DELLA_STRUTTURA.md. The checks cited are in these files:
- the fast pairs (4.4): lettura_locale/coppie/centrifuga.py;
- the total field (4.5): lettura_locale/campo/specchio_campo.py and lettura_locale/campo/valuta_totale.py;
- the mass ratio (4.5): lettura_locale/coppie/rapporto_masse.py;
- the Gaia field (4.5): lettura_locale/campo/sensibilita_campo.py;
- the screening readings (4.5): documenti/PROTOCOLLO_LIVELLO2.md and NOTA_ESITO_LIVELLO2.md;
- the direction (4.6): the folder lettura_locale/direzione;
- light (4.7): the folder lettura_locale/luce; the red and blue lenses: lettura_locale/luce/lenti_rosse_blu.py;
- galaxies in their environment (4.8): the folder lettura_locale/galassie; the five curves: lettura_locale/galassie/cinque_curve.py;
- systems without rotation (4.9): the folder lettura_locale/sistemi; the test with Y★ changed: lettura_locale/sistemi/sistemi_sensibilita.py.
- **Section 5, the registrations for DR4:** lettura_locale_dr4/; k5c_dr4/ for the instantaneous form; lettura_locale/direzione/ for the direction.
- 5.1: the pipeline programs of MM 2.56 are copied with exact, listed substitutions, in the *.diff files. A pipeline is a chain of programs that work one after the other.
- 5.3: the note on noise is documenti/NOTA_RUMORE_DR3.md. The registered protocol and programs are in lettura_locale/rumore.

- 5.4: the rule over several series is lettura_locale/rumore/AGGIUNTA_DR4_PIU_SERIE.md. The programs are in dr4_programmi/. esegui_dr4_serie_b.py runs the 190 simulations. valuta_dr4_serie_b.py applies the rule. The MM 2.56 validity conditions it checks are: V1 data, V2 libraries, grid and data fingerprints, V4 fit.
- 5.5: the anisotropy protocol is in lettura_locale/direzione: BOZZA_PROTOCOLLO_ANISOTROPIA_DR4.md and AGGIUNTA_1. Addendum 2 is lettura_locale/direzione/TRADUZIONI_E_AGGIUNTA_2_DR4.md. The code is dr4_programmi/anisotropia_dr4.py.

The commands of 2 December 2026 (section 5.6). After the simulations of MM 2.56 and of the two local forms:

- valuta_lettura.py and valuta_k5c_c1.py in dr4 mode;
- esegui_dr4_serie_b.py <DR4 data> <output> tutto, then valuta_dr4_serie_b.py <DR4 data> <output> dr4 (section 5.4);
- anisotropia_dr4.py <DR4 data> <output> dr4 (section 5.5).

The verification. The work was checked in several rounds by Codex/GPT-6, an AI assistant.

- **The first rounds.** Codex/GPT-6 checked Levels 1–2 (documenti/verifica_astra/RAPPORTO_GIRO1.md). Level 1 is the chain taken to the Sun (section 3). Level 2 covers the screening readings (section 4.5). It also checked the instantaneous registration for DR4 (k5c_dr4/documenti/verifica_astra/RAPPORTO_K5C.md).
- **The third round** checked the full local reading (lettura_locale_dr4/documenti/verifica_astra/RAPPORTO_LETTURA.md). It confirmed the dry-run numbers and the exact reproduction of MM 2.56.
- **The changes.** Changes of wording and documentation were applied (the CORREZIONI_* files). No arithmetic error changing an outcome was found.
- **The fourth round** checked the additions made after the second version: sections 4.3–4.9 and 5.3–5.5 (documenti/verifica_astra/RAPPORTO_PARTI_NUOVE.md).
- Two remarks did not hold. The first was a formatting error in dipolo_dominante.py: tuple addition concatenates them, that is, it puts them in a row. The second was an unchecked star order. In fact star 1 is the brighter in all rows of both catalogues.
- The other remarks were applied:
- the w fit with the full covariance;
- numeric rules for the anisotropy protocol (addendum 3);
- the test with Y★ changed and a comparison at the same gravitational state for dwarfs and clusters;
- the wording on S2, on the fast pairs, on the lenses and on the mass ratio;
- a recount of the SPARC concentration.
- No outcome changed (documenti/CORREZIONI_DOPO_ASTRA_V3.md).
- **The fifth round** checked the additions made after the fourth: the total field, the red and blue lenses, dwarfs and clusters, and the programs for DR4 (documenti/verifica_astra/RAPPORTO_QUINTO_GIRO.md).
- The corrections applied (documenti/CORREZIONI_DOPO_ASTRA_V4.md) are these:
- the profile fit with the correct model of the radius error;
- the partial correlation computed as Pearson on rank residuals. The partial correlation is the link between two quantities when a third is held fixed;
- the dwarf comparison without the gas-rich systems;
- P3 without the Magellanic Clouds, which are two satellite galaxies of the Milky Way;
- validity and completeness checks in the programs for DR4 (versions _b);
- softened causal wording (generations, tide, satellites), that is, less certain sentences about what causes what;
- "3/2 is the arithmetic mean of 1 and 2".
- Two remarks were checked and not applied. The first concerns the meaning of rhalf_sph_physical, a column of the catalogue: our M_dyn equals the catalogue's Wolf mass within 0.6 %. The second concerns the counts of measured metallicity spreads: they are 38, as written.
- No outcome changed.

The registrations. Every protocol and every program was registered before it was run, with a SHA-256 fingerprint and the UTC time.

- The *fingerprint* is a code computed from the file. If the file changes even by a comma, the fingerprint changes.
- UTC is the reference time for the whole world.
- The registers are included. REGISTRO_estratto.txt contains the registration lines.
- Some files differ from their registered originals only in the text lines listed in CAMBI_TESTO.md. They are five programs, the noise protocol and addendum 2. The code fingerprints are identical.

What is in the package.

- `codice/` and `risultati/`: Level 1 (passo123.py, passo4.py) and Level 2 (livello2.py, livello2_c1.py), with logs and JSON. A *log* is the diary that a program writes while it works. JSON is a file format for data.
- `codice/esterni/`: the earlier verification code and the MM 2.56 solver, unchanged.
- `lettura_locale/`: the checks of section 4, with logs and JSON:
- premessa_mcd.py and premessa_inerzia.py;
- corpo_dominante/, coppie/, campo/ (with the 10 simulations of the total-field test), direzione/, luce/, galassie/ and sistemi/;
- rumore/ (sections 5.3–5.4), with the 80 simulations of the noise test and their SHA-256.
- `lettura_locale_dr4/` and `k5c_dr4/`: the registrations for DR4. They contain protocols, copy generators and diffs, programs, logs, and the 10 + 10 simulations on DR3 with SHA-256. The orbit libraries (about 210 MB each) are not included. They regenerate in about six minutes from seeds 18 and 17.
- `dr4_programmi/`: the registered programs for DR4 (the anisotropy; the rule over several series), their dry runs on DR3, the identity check and the logs of the two new libraries. The libraries are not included: they regenerate from seeds 23 and 24.
- `dati/`: the MM 2.56 sample, the JPL page, the source list.
- `documenti/`: inventories, protocols, outcome notes, the structure of the local reading and its checks, the note on simulation noise, the verification reports, the registers.
- The programs contain the author's paths (E:...), which have to be adapted.

This version. This student version was written with an AI assistant (Claude) from the archived text in English. First a full Italian translation was made. Then the translation was rewritten for a school student. The numbers were checked with a program: the translation against the English text, this version against the translation. The archived text is the reference. This English text is a translation of the Italian student version, which the author read and approved.

Glossary

- **a_0 (a-zero).** A tiny acceleration. Below a_0, in MOND, gravity is stronger than Newton's. In the chain it is $cH_0/6$.
- **Acceleration.** How fast the speed changes. It is measured in metres per second squared (m s^{-2}).
- **Anisotropy.** An effect that changes with direction.
- **Anomaly.** The part of gravity in excess of Newton.
- **au (astronomical unit).** The mean distance between the Earth and the Sun.
- **Boost (ν, η).** The number by which Newton's gravity is multiplied. ν is the law that gives it; η is its value for a pair.
- **χ^2 (chi-squared) and $\Delta\chi^2$.** χ^2 measures how far a model is from the data: the smaller it is, the better. $\Delta\chi^2$ is the difference between two χ^2.
- **Control pairs and low band.** The control pairs are close, with high acceleration. The low band contains the 919 low-acceleration pairs.
- **Covariance.** How much the errors of two measurements go together.
- **Curvature.** How much a field changes from one point to another. A uniform field has no curvature.
- **Deviance.** A number that measures how far a hypothesis is from the data, like χ^2.
- **dex.** A way of measuring ratios in powers of ten. 1 dex is a factor of ten; 0 dex means equal.
- **Dipole and quadrupole on the sky.** The dipole (cos φ) makes a difference between one side and the opposite side. The quadrupole (cos 2φ) is the same on the two opposite sides.
- **Dominant, compact, extended body.** Dominant is the body whose local MCD prevails. It is compact if it is small compared with the distance of the relation, like the Sun. It is extended if it is spread along the relation, like a galaxy.
- **Dry run.** A rehearsal on data already seen, to check that everything works.
- **Dwarf spheroidal and ultra-faint galaxies.** Small ball-shaped galaxies. The ultra-faint ones are the smallest and the least bright.
- **Dynamical mass (M_dyn).** The mass obtained from the motion of the stars.
- **Eccentricity.** How flattened an orbit is: 0 is a circle, near 1 it is a very elongated ellipse.
- **Environment field.** The gravity of nearby galaxies, felt by a galaxy.
- **External field.** The gravity that comes from outside a system. For the Solar System it is that of the Milky Way.
- **Field.** It says how much and in which direction gravity pulls, at every point of space.
- **Forward model.** A computation that starts from the law of gravity and predicts what should be seen in the data.
- **Gaia and its data releases.** Gaia is a satellite that measures the position and the motion of stars. DR3 is its data release already published. The next release is called DR4.
- **General relativity.** Einstein's theory of gravity.
- **Global-field reading.** The reading of the chain: the field of the Galaxy enters all of space through the field equation, down to the Sun. Cassini excludes it.
- **Globular cluster.** A ball of old stars, much smaller than a galaxy. Its stars were born nearly together.
- **Gravitational state (x_h + e_N).** The sum of the interior gravity of the system and the external field, divided by a_0.

- **Inertial state.** The rule for relations without a dominant body: the reading is the time average along the orbit.
- **Intrinsic scatter.** The spread of the data around a law that does not come from the measurement errors.
- **Likelihood and log-likelihood.** The likelihood says how probable a hypothesis makes the observed data. The log-likelihood is its logarithm. The higher it is, the better.
- **Local reading.** The reading proposed in this record: the anomaly lives in the relation between two bodies, with six rules.
- **MCD.** The greatest common divisor. In the MM framework it is the common unit that a relation counts. In this record the local MCD of a mass is R_s/r.
- **Median.** The value that sits in the middle: half of the cases are below, half above.
- **Metallicity.** The amount of elements heavier than helium in a star. If it changes from star to star, the stars are of different generations.
- **Mirror.** In the MM framework, the name of what is usually attributed to dark matter (the text says *dark-matter mirror*). A system carries a mirror when its dynamical mass is more than three times the visible mass: B = M_dyn/M_b > 3.
- **MOND, QUMOND, AQUAL.** MOND is Milgrom's idea: below a_0 gravity is stronger than Newton's. QUMOND and AQUAL are two versions of MOND written with a field equation.
- **MOND radius (r_M).** The distance at which the Newtonian acceleration of a body drops to a_0. For the Sun it is 7,371 au.
- **Newtonian.** Following Newton's law of gravity, with no boost.
- **Noise and seed.** The seed is the starting number of the random numbers of a simulation. The noise is how much the result changes when the seed changes.
- **Oort cloud.** A large cloud of comets at the edges of the Solar System.
- **Orbit library.** A large set of orbits computed on a computer for the simulations.
- **Outcome.** The result of a test. A negative outcome makes a reading fall, not the MM framework.
- **parsec, kpc, Mpc.** The parsec (pc) is a unit of distance used for stars. A kpc is a thousand parsecs, a Mpc a million parsecs.
- **Phantom density.** In QUMOND, the invisible mass that seems to be in empty space. It is a tool of the equation, not an object.
- **Protocol.** The list of the steps of a test, written before doing it.
- **Quadrupole (Q_2).** A deformation that stretches along one direction and squeezes at the sides, like the tide. Cassini measures its value around the Sun.
- **QUMOND chain.** The row of four choices of MM 2.56: $a_0 = cH_0/6$, the law ν, the Milky-Way field V^2/R, the QUMOND equation.
- **Radial acceleration relation.** An observed law in disc galaxies. It links the measured acceleration to the acceleration of visible matter alone.
- **Radio tracking.** Following a probe from Earth with radio signals, to know its position.
- **Reading, read.** In MM a body reads a field when its motion takes it into account. A reading is a rule of this kind: an interpretation, which can be put to the test.
- **Registered.** Written and fixed with its date before seeing the data, so that it cannot be changed afterwards.
- **Relation.** The link between two bodies that attract each other, like the Sun and a planet, or the two stars of a pair.
- **Rotation curve.** How fast stars and gas go round, at each distance from the centre of a galaxy.
- **Satellite.** A small galaxy that goes round a larger galaxy.
- **Schwarzschild radius (R_s).** The radius that a black hole with a given mass would have.
- **σ (sigma) and z.** σ is the typical error of a measurement. z says by how many σ a prediction lies away from the measurement.
- **Standard deviation.** How far the values lie, on average, from their mean.
- **Systems without rotation.** Systems in which the stars move in all directions, like the bees of a swarm.
- **Tensorial alignment.** The way in which the direction enters the local reading. What counts is the square of the cosine of the angle between the pair and the Galactic centre: $\cos^2\theta$.
- **Total field and baryonic field.** The baryonic field comes from ordinary matter alone. The total field also includes the dark-matter mirror.
- **Velocity dispersion.** How different the velocities of the stars of a system are from one another.
- **Weak lensing.** A small deformation of the images of distant galaxies, due to the gravity of nearer galaxies.
- **Wide binaries.** Pairs of stars that go round each other, very far apart.
- **x and x_eff.** x is Newton's acceleration divided by a_0. x_eff is its average along the orbit.

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